



Roll no. 124111003

Department of Physics
National Institute of Technology Kurukshetra

Course: B.Tech (All Branches)

Exam: END SEMESTER

Date: 04-12-2024

Subject: Engineering Physics

Course Code: PHIC 101

Duration: 3 hours

Note:

- Attempt Five questions in all.
- Do not write anything on the question paper, except your Roll no.
- Attempt each part of a particular question at the same place in your answer sheet.
- Wherever necessary, draw relevant, neat and labelled schematic diagrams.

Relevant Physical constants:

$h = 6.67 \times 10^{-34} \text{ J.s}$; $c = 3.0 \times 10^8 \text{ m/s}$; $e = 1.6 \times 10^{-19} \text{ C}$; $m_e = 9.1 \times 10^{-31} \text{ kg}$; $1 \text{ eV} = 1.6 \times 10^{-19} \text{ Joule}$

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|----|---|
| 1. | (a) What are the differences between matter waves and electromagnetic waves? [2]
(b) What is Poynting vector. The average intensity of noontime sunlight on a clear day is about 2 kW/m^2 . what area of 50% efficient solar collectors is needed to replace a 5 kW water heater? [3]
(c) What is relaxation time? Find the relaxation time of conduction electrons in a metal of resistivity, $1.54 \times 10^{-8} \Omega\text{m}$, if the metal has 5.8×10^{28} conduction electrons per m^3 . [3]
(d) What is relativistic mass?. How does it change with velocity? [2] |
| 2. | (a). Derive the energy spectrum and wavefunctions for a particle confined in a one-dimensional box, and demonstrate that the energy levels are quantized, in contrast to classical mechanics. Plot the first three wavefunctions and their corresponding probabilities. [7]
(b). If the potential inside the box is constant but nonzero, how would the wavefunctions and energy levels be affected? [3]
<p style="text-align: center;">OR</p>
(a). Discuss the significance of the displacement current term in Maxwell's equations and its role in predicting electromagnetic waves. [5]
(b). A plane electromagnetic wave is propagating in the +z-direction with the electric field given by, $\mathbf{E}(z, t) = E_0 \exp[i(kz - \omega t)]\hat{x}$. Derive the corresponding magnetic field and determine the wavelength and frequency of the wave if $E_0 = 10 \text{ V/m}$ and $k = 10 \text{ m}^{-1}$. [5] |
| 3. | (a). What is Packing fraction of the unit cell? Calculate the maximum packing fraction of the unit cell volume that can be filled by hard spheres in the BCC and FCC crystal structures? Which structure most efficiently fills the space? [5]
(b). What are Miller indices, why are they represented as integers, and what is their significance in crystallography? Also, find the Miller indices for a plane parallel to the z-axis and cutting the x- and y-axes at $a/2$ and $b/3$, respectively. [5] |

OR

- (a). Explain the Fermi-Dirac distribution function and its role describing the behavior of fermions within a system. How does this distribution behave as the temperature approaches absolute zero, and what are the consequences for the electronic properties of solids, such as conductivity and heat capacity? [5]
- (b). Consider a semiconductor at room temperature ($T=300\text{ K}$) with a Fermi energy $E_F=0.5\text{ eV}$ and an energy gap of 1.1 eV . Using the Fermi-Dirac distribution, calculate the probability that an electron occupies a state at the conduction band edge ($E = E_F + 0.5\text{ eV}$). [5]

4.

- (a). What is stimulated emission? Discuss how does the principle of stimulated emission differ in semiconductor lasers compared to He-Ne lasers with neat diagrams? [5]
- (b). A step-index optical fiber has a core refractive index of 1.45 and an acceptance angle of 8.50° in air. If the fiber is immersed in water (with a refractive index of 1.33), determine the change in numerical aperture and acceptance angles. [5]

5.

- (a). Derive the expressions for length contraction and time dilation, using appropriate diagrams, as described in the Special Theory of Relativity. [5]
- (b). We consider a spaceship that moves away from earth with the velocity $v = 0.866c$. It emits two light signals spaced by $\Delta t' = 4\text{ s}$. Here c is velocity of light.
- What is the time distance ΔT (earth time) between the two signals arriving on earth?
 - What distance, measured from earth, did the spaceship cover between emitting the two signals?
 - A body at rest in the spaceship has the mass $m_0 = 1\text{ kg}$. What is its kinetic energy measured from earth? [5]

OR

- (a). Explain the concept of neutron cross-section and derive its mathematical expression. [5]
- (b). What is the significance of the critical size of fuel? How it affects the controlled chain reaction mechanism? [5]

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